Warren Lin

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Education

San Jose State University — San Jose, CA
B.S. Industrial Systems Engineering (intending to transfer into Electrical Engineering) — *2026 – 2030*

- Participating in the Software and Computer Engineering Society (SCE) and Chip Design Initiative (CDI) at SJSU.
- Relevant Coursework: CS46A

Experience

Business & Electrical Lead — Danville, CA
FRC Team 1280, Ragin' C Biscuit — *2024 – 2026*

- Grew sponsorship revenue **34%** year-over-year against a **$50,000** club budget, by writing and submitting **50+** grant and corporate sponsorship applications.
- Kept the team on schedule for **3 seasons**, building a Notion-based project management system to track corporate deadlines and funding targets.
- Onboarded **30** new members onto the electrical subteam, delivering hands-on electrical safety and systems training.
- Cut electrical debugging time by **40+ hours**, by designing a CAN Bus error redundancy and isolation system for in-season maintenance.
- Bridged the gap between software and hardware, developing a **4+ camera** AprilTag localization setup on a Linux-based Mac Mini.

Contributor — Cambridge, MA (Remote)
MIT Open Compute Lab — *2025 – Present*

- Implemented and tested the **load-store unit** for an open-source **RV32I** core in SystemVerilog; **5** tests passing in **Verilator**.
- Informed the lab's roadmap for an **RV32I processor**, auditing **14** RISC-V conference talks and presenting findings and proposed extensions to the research group.
- Extended Linux compatibility for the lab's Apple M1-class SoC initiative, by getting **closed-source x86 applications running on aarch64** through binary translation.

Projects

Ichika – *CI for Remote HDL Synthesis | Nix, Vivado* — Summer 2026

- Built a Nix flake that offloads Vivado HDL synthesis to a remote server with Vivado pre-installed, letting contributors skip local toolchain setup.
- Adopted lab-wide by the MIT Open Compute Lab (**136** members).

HackMIT – 1st Place | *First Overall, Software Design Track* — Fall 2025

- **Colmena** – won 1st place among **1000+ undergraduates** by building a **map-based application designed around human mobility**, using **SvelteKit, Haskell, and MentraSDK**.

Stock-Prediction RL Agent | *Berkeley CCE, Deep Q-Learning* — Summer 2025

- Trained a Deep Q-Learning agent on a 15-stock environment; **outperformed a buy-and-hold baseline** on held-out data.
- Diagnosed and documented training-data overfitting in the model, tracing the cause to unmodeled **business cycles and significant market events**.

jade – *Just Another Display Engine | HDL, Modal Logic, GPUs* — Ongoing

- Specifying a graphics-first ISA for an open GPU with cores built on modal-logic morphisms; spec covers **268 instructions** across **50** opcode families.
- **wavefront-like architecture** implemented in **Chisel**, putting graphical operations as primitives and offloading non-time-critical work to software.

Athena – *Critical Infra for functor.systems | Linux, NixOps, Docker* — *active since* Winter 2024

- Maintain a high-end workstation as a server with **greater than 90%** reliability
- Host Docker containers and other daemons for functor.systems

Technical Skills

Languages: Python, Nix, Rust, C/C++, Bash, SystemVerilog, Java(FRC), Chisel/Scala, some Assembly (ARM32, RV32)
Embedded/Hardware: Arduino, Raspberry Pi, NXP, STM32, Xilinx FPGA, HDL development, GTKWave, Verilator
Tools: Linux, Git/GitHub, KiCad, Vivado, LaTeX, SSH